

Retail Forecasting

Final Deliverable — IEOR 4578

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- ▶ Welcome to the final project presentation for **IEOR 4578**.
- ▶ This project features an end-to-end machine learning pipeline for forecasting a 12-week retail demand horizon across the Top 1,000 SKUs of a UK-based online retailer.
- ▶ This presentation will walk you through our complete methodology: from triple-clustering (Behavioral, Semantic, and Volumetric) and SOTA model training, all the way to the deployment of an Auto-Routing LLM Agentic Assistant.



Executive Summary: Strategic Operational Impact

- ▶ **Architectural Overhaul:** Completely refactored the forecasting pipeline from a monolithic, error-prone notebook to a modular, production-ready repository. We significantly expanded the model library, implemented global hyperparameter optimization via Optuna, and integrated rich exogenous signals such as pricing elasticity and holidays.
- ▶ **Main Results:**
 - ▶ –14.2% reduction in **Global WMAPE** compared to the previous team's baseline.
 - ▶ –91.7% reduction in **Computational Runtime** (from 60 minutes to < 5 minutes cumulatively).



Agenda

1. **Business Context & Objective**

Retail dynamics, Pareto skew, and the cost of Stockouts vs. Overstock

2. **Project Evolution & Methodology Upgrades**

Missing Components, Errors, and Architectural Overhaul

3. **Data Strategy & Engineering**

Value Creation, New Predictive Features, and Triple-Clustering

4. **Modeling**

Strategy, LightGBM, NS-Transformer, Ridge, SARIMAX, and Prophet

5. **The Agentic Layer**

LangGraph Framework, LLM Auto-Routing, and Business Insights

6. **Results & Evaluation**

WMAPE Benchmarking, Transformer Peak Specialization, and Trade-offs

7. **Improvements & Next Steps**

Computational Overhead Reduction and Strategic Roadmap



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Business Context & Objective: Retail Dynamics

Inventory managers must plan procurement weeks in advance to satisfy a highly volatile global demand.

Operational Mechanics:

- ▶ Transactional data is aggregated into **Weekly Cycles** (Monday-to-Sunday).
- ▶ Demand is highly **discrete and intermittent**: many SKUs experience long periods of zero sales followed by sudden bulk orders.
- ▶ Decisions are made on a **12-week horizon** (one full fiscal quarter).

The Challenge: extreme **Pareto Skew** (0.4% of SKUs drive the majority of volume), making precise blockbuster prediction a financial imperative.



Business Context & Objective: Inventory Risks

When actual sales deviate from the 12-week forecast, the business faces severe financial erosion through two symmetric risks.

Over-forecasting (Overstock)

Purchased too much inventory.
Capital is tied up in stagnant stock. Incurs high **holding costs** and forces aggressive **markdowns/losses**.

Under-forecasting (Stockout)

Purchased too little inventory.
Immediate **lost revenue**. Leads to customer dissatisfaction and B2B wholesale **churn**.

Our Value Proposition: By deploying a cluster-aware MLOps pipeline, the retailer can optimize safety stock, protect profit margins, and handle the "Cold-Start" problem through LLM-driven semantic context.



Business Context & Objective: Project Scope

Build and compare machine learning models capable of predicting the next **12 weeks** of physical unit demand for the Top 1,000 revenue-generating SKUs.

Approaches evaluated:

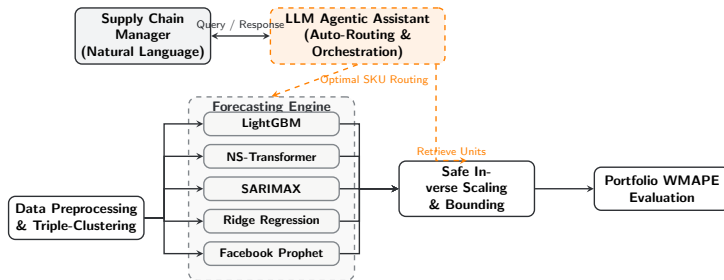
1. **LightGBM**: Specialized for zero-inflated demand.
2. **NS-Transformer**: SOTA Deep Learning for non-stationary peaks.
3. **SARIMAX & Ridge**: Baselines for linear memory and lags.
4. **Facebook Prophet**: For macro-seasonal trend decomposition.

Triple-Clustering: segmenting the catalog by *Behavioral Shape*, *Semantic Identity* (Gemini LLM), and *Volume Tier* (Jenks).

Agentic AI Assistant: A LangGraph-based interface with Auto-Routing to trigger the mathematically optimal model per SKU.



Business Context & Objective: System Architecture



- ▶ **Modular Pipeline:** Seamless flow from Triple-Clustering to Inverse Scaling.
- ▶ **Optimal Routing:** LLM selects the best model per SKU based on WMAPE.



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Project Evolution & Methodology Upgrades: Missing Component 1

Assignment Requirement

“...cluster consumers on their usage patterns, or products in similar groups... and then build models for each cluster.”

- ▶ **What was implemented:** The baseline executed a monolithic for loop over 250 SKUs, training completely isolated and independent models.
- ▶ **The Gap:** While the team calculated the Coefficient of Variation (CV) to categorize SKUs, it was used solely for retrospective appendix diagnostics, never for algorithmic clustering. The models entirely missed the opportunity to learn shared cross-SKU seasonal patterns, inflating computational time.



Project Evolution & Methodology Upgrades: Missing Component 2

Assignment Requirement

“...create time-based features, incorporate holidays or other schedules, and add external data...”

- ▶ **What was implemented:** The data preprocessing aggregated only historical Quantity and Revenue, feeding a purely univariate time-series into the algorithms.
- ▶ **The Gap:** By omitting UK national holidays, pricing elasticity, and promotional flags, the models remained completely “blind” to external retail drivers. This directly contributed to massive forecasting errors during peak holiday shopping periods.



Project Evolution & Methodology Upgrades: Missing Component 3

Assignment Requirement

“...ask the LLM to provide you with the latest forecasts. The LLM will sort of run a mapping...”

- ▶ **What was implemented:** The codebase revealed a simple Python function relying on hardcoded Regular Expressions (`re.findall`) to perform exact string matches against a predefined list.
- ▶ **The Gap:** There was no actual Large Language Model (LLM) instantiated, no natural language reasoning, and no dynamic tool-calling involved. It was merely a standard search bar masquerading as an AI agent.



Project Evolution & Methodology Upgrades: Errors:

- ▶ **Terminological Contradiction:** The baseline claimed "SARIMA" was their strongest model. However, their code explicitly set `seasonal_order=(0,0,0,0)`, reducing it to a standard non-seasonal ARIMA(1,1,1). In retail forecasting, seasonality is the most critical component. Because their model had no seasonal memory, the multi-step autoregressive forecasts quickly collapsed into a flat, constant mean. Their final "12-week forecast" was effectively a 2-week forecast padded with identical flat values.
- ▶ **Silent Model Fallbacks:** The baseline used broad `try...except` blocks. If a model failed or produced NaN, it silently overwrote the prediction with a Naïve model. This corrupted the evaluation metric, masking true algorithmic failures from stakeholders.



Project Evolution & Methodology Upgrades: Architectural Overhaul

Our Architectural Overhaul:

1. **Exogenous Features:** Injected pricing elasticity and explicit UK holidays to capture dynamic market volatility.
2. **Cluster-Based Modeling:** Shifted to multi-dimensional clustering, training global models per cluster to capture shared seasonalities and drastically reduce computational overhead.
3. **SOTA Architectures:** Deployed LightGBM, NS-Transformer, and Ridge Regression to ensure robust predictive accuracy.
4. **True Agentic Layer:** Implemented LangGraph orchestration with dynamic Auto-Routing to provide decision support.
5. **Robust Metrics:** Transitioned to WMAPE for true business impact, ensuring a volume-weighted financial evaluation.



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Data Strategy & Engineering: Value Creation

From: A raw event-driven transactional database (≥ 1 M rows), heavily noisy and conflated with reverse logistics.

To: A strictly chronological, machine-learning-ready Parquet dataset enriched with pricing dynamics, historical lags, and semantic embeddings for the Top 1,000 SKUs.

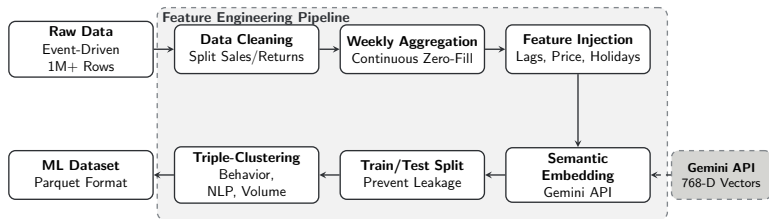


Figure: The end-to-end data transformation pipeline for the Retail dataset.

Calendar & Exogenous Signals

- ▶ **Cyclical Encodings:** Sine and Cosine transformations for continuous time.
- ▶ **Holidays & Events:** UK National Holidays and an explicit `is_christmas_window` flag to isolate the Q4 shopping surge.

Pricing Dynamics

- ▶ **Promotional Elasticity:** A categorical `is_on_promotion` flag triggered when the weekly price drops $< 85\%$ of the historical median.

Autoregressive & Static Profiling

- ▶ **Lags & Returns:** Strictly shifted rolling statistics and a `return_rate` flag to identify defective batches.
- ▶ **Syntetos-Boylan Metrics:** ADI and CV^2 calculated *exclusively* on the training set to quantify sparsity without data leakage.



Data Strategy & Engineering: Seasonal Behavior

- ▶ **Goal:** Capture temporal rhythms independent of sales magnitude.
- ▶ **Method:** K-Means on *L2-Normalized* 52-week sales profiles.
- ▶ **Key Archetypes:** *Pre-Christmas Peak*, *Spring Surge*, *Autumn Flash*, and *Stable Base*.

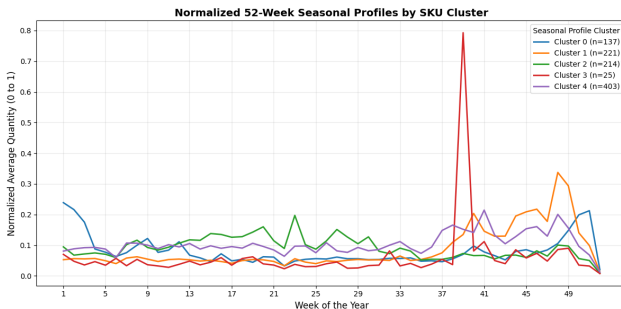


Figure: Normalized demand shapes: isolating seasonality from volume.



Data Strategy & Engineering: Semantic Identity

- ▶ **Goal:** Provide context for "Cold-Start" products via LLM metadata.
- ▶ **Method:** 768-D vectors via Google Gemini Embedding 2 + UMAP reduction.
- ▶ **Result:** Grouping by functional similarity (e.g., *Christmas Decor*, *Enamel Mugs*, *Jewelry*).

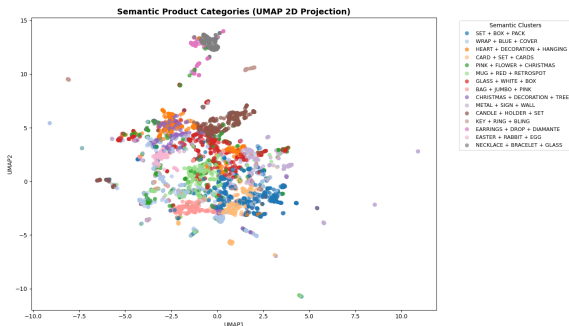


Figure: UMAP projection of Gemini Embeddings.



Data Strategy & Engineering: Volumetric Scale

- ▶ **Goal:** Address extreme Pareto skew (0.4% of SKUs = 50k+ units).
- ▶ **Method:** Jenks Natural Breaks to find objective "cliffs" in popularity.
- ▶ **Impact:** Prevents over-forecasting slow-movers by locking them into a "Low-Volume" prior.

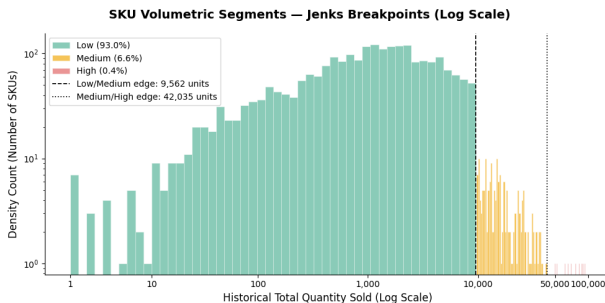


Figure: Jenks breaks: identifying natural popularity tiers.



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- ▶ **Chronological Split:** 12-week out-of-sample test horizon (Q4 2011) to rigorously evaluate robustness during the extreme Black Friday/Christmas peak.
- ▶ **Cluster-Aware Training:** Training one dedicated model per behavioral cluster. Increases signal-to-noise ratio and drastically reduces computational overhead.
- ▶ **Global Hyperparameter Tuning:** Using Optuna across the full dataset to optimize hyperparameters globally, preventing overfitting on small or highly volatile segments.
- ▶ **Dynamic Model Selection:** An "Auto-Router" evaluates out-of-sample predictions and statically assigns the mathematically optimal model to each specific SKU.



Modeling: LightGBM

- ▶ **Architecture:** Tree-based gradient boosting. Natively evaluates categorical splits (volume tier, NLP cluster) via histograms, avoiding matrix inflation.
- ▶ **The Tweedie Distribution:** Retail demand is highly zero-inflated and Pareto-skewed. The tweedie objective perfectly bridges Poisson (zeros) and Gamma (continuous) distributions.
- ▶ **Optuna Tuning:** Optimized variance power to ≈ 1.77 and enforced strict leaf constraints to prevent learning anomalous histories.
- ▶ **Result:** The undisputed global leader. Highly aggressive and capable of predicting sudden vertical promotional spikes from a zero-baseline.



Modeling: Non-Stationary Transformer

- ▶ **Core Innovation (NeurIPS 2022):** Standard normalization destroys temporal variance. This architecture solves "over-stationarization" via **De-Stationary Attention**.
- ▶ **Mechanism:** Utilizes Projector MLPs to learn scaling (τ) and shifting (Δ) factors directly from un-normalized statistics, restoring predictive variance inside the attention map.
- ▶ **Tensor Construction:** Sliding windows map 16 weeks of context into a 12-week forecast horizon, compressed via $\log_{1p}$ to stabilize extremes.
- ▶ **Result:** Exhibits a brilliant progressive learning curve. It achieves its lowest error precisely during the highly chaotic Q4 holiday window.



Modeling: SARIMAX

- ▶ **Stochastic Modeling:** Classical econometrics explicitly capturing historical memory and exact 52-week seasonal cycles via exogenous anchoring.
- ▶ **Optuna Configuration:** Converged on Order **(0, 2, 1)** and Seasonal Order **(1, 0, 1, 52)**, revealing non-linear trend acceleration through double differencing ($d = 2$).
- ▶ **Reactive Error Mapping:** By setting $p = 0$ and $q = 1$, the model prioritizes correcting its trajectory based on recent forecast errors rather than absolute past sales inertia.
- ▶ **Structural Limits:** Despite its mathematical rigor, the rigid linear assumptions fail to accurately map the discontinuous and spiky nature of the intermittent retail catalog.



Modeling: Ridge Regression

- ▶ **Architecture:** Multiple Linear Regression utilizing L2 regularization to penalize excessive coefficient weights.
- ▶ **Feature Engineering:** Requires heavy preprocessing: rigorous One-Hot Encoding for categories, MinMax scaling for ratios, and $\log_{1p}$ scaling for volumes.
- ▶ **Optuna Tuning:** Globally optimized penalty ($\alpha \approx 99.32$) to mitigate the severe multicollinearity inherent in highly correlated autoregressive lags.
- ▶ **Result:** A remarkably stable baseline. It tightly anchors predictions to recent autoregressive inertia, maintaining flat variance across the entire 12-week horizon.



Modeling: Facebook Prophet

- ▶ **Architecture:** Additive macro-seasonal decomposition. Configured with `yearly_seasonality=True` and explicit UK national holidays.
- ▶ **Exogenous Regressors:** Anchored to commercial realities by injecting `is_on_promotion` flags and pricing elasticity vectors.
- ▶ **Optuna Tuning:** Optimized the trend flexibility (`changepoint_prior_scale` ≈ 0.114) to balance underfitting and extreme variance.
- ▶ **Result:** Severely underperforms on the retail catalog. Its underlying continuous Fourier curves inherently smooth over isolated promotional spikes and sudden stock-outs.



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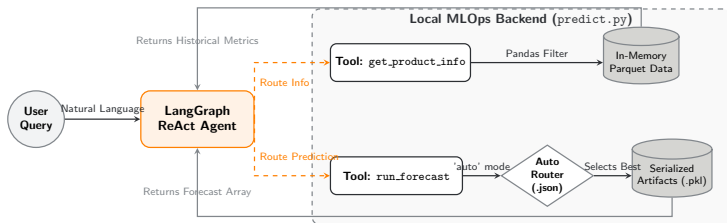
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The Agentic Layer: LangGraph & ReAct Framework

- **Objective:** Abstract complex MLOps (clustering, inverse scaling, auto-routing) into a frictionless natural language interface for Supply Chain Managers.
- **ReAct Paradigm:** The LLM follows a *Reasoning and Acting* loop to solve intents dynamically using external Python tools.



The Agentic Layer: Tool Integration and Context Retrieval

1. Context Retrieval (`get_product_info`)

- ▶ Sub-second access to the master Parquet dataset.
- ▶ Translates numerical clusters into actionable business profiles (e.g., "*Winter/Christmas Peak*").

2. Live Inference (`run_forecast`)

- ▶ Reads a `best_model_per_sku.json` dictionary to dynamically trigger the mathematically optimal algorithm per product.
- ▶ Loads specific `.pkl` weights and performs on-the-fly inverse scaling (`expm1`) and zero-bounding.



The Agentic Layer: Business Interpretation

From Raw Arrays to Strategic Intel: The Agent's system prompt enforces a strict response hierarchy:

- ▶ **Executive Summary:** Rapid overview of total volume and peak metrics.
- ▶ **Analyst Interpretation:** Combines the numerical forecast with semantic/seasonal context to suggest direct inventory actions.

Example Interaction (Product 85123A)

Manager: "Forecast the demand for product 85123A for the next 8 weeks. Please use the 'auto' mode..."

Agent: "I have generated a demand forecast for product 85123A using the 'auto' model selection, which chose **LightGBM (LGB)** as the most accurate model.

- **Total Sales Forecast:** 4,410 units
- **Average Weekly Sales:** 551.3 units/week
- **Peak Expected:** 649.0 units (week of October 10, 2011)"



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1. The Economic Standard: Portfolio WMAPE

$$\text{WMAPE} = \frac{\sum |\text{Actual} - \text{Forecast}|}{\sum \text{Actual}} \times 100$$

Pools absolute errors over total volume. Directly reflects the financial risk of overstock/stockouts for warehouse managers.

2. The Diagnostic Standards: Median MAPE & MAE

Median MAPE filters out zero-demand noise to evaluate item-level "shape" accuracy. **MAE** expresses the average error in exact physical units.



Results & Evaluation: Global Performance Benchmark

Model Architecture	WMAPE	Med. MAPE	MAE (Units)
LightGBM	57.13%	52.46%	41.44
NS-Transformer (SOTA)	60.81%	55.86%	43.53
Ridge Regression (Baseline)	60.99%	56.54%	44.24
Facebook Prophet	89.37%	78.46%	64.82
SARIMAX	90.28%	79.61%	65.49

- ▶ **Global Winner:** LightGBM. The Tweedie objective perfectly conquers the zero-inflated nature of the retail catalog.
- ▶ **Robust Baselines:** The NS-Transformer and Ridge Regression perform exceptionally well, drastically outperforming classical models.



Results & Evaluation: Key Architectural Insights

1. The Transformer's Peak Specialization Temporal analysis shows the NS-Transformer's error *drops* to an unmatched 54.68% during the chaotic Black Friday/Christmas window. De-Stationary Attention is uniquely suited for capturing extreme anomalies.

2. The Cluster 3 Anomaly & Auto-Routing Validation

LightGBM fails on Cluster 3 (highly erratic Autumn flash peaks), misinterpreting the sudden spikes. To overcome this, our Agentic Auto-Router dynamically assigns these volatile SKUs to the Transformer or Ridge models.

3. The Classical Statistical Collapse Prophet and SARIMAX mathematically break down. Continuous Fourier curves and rigid ARIMA matrices are structurally incompatible with intermittent retail sparsity.



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Improvements & Next Steps: Architectural Evolution

Overcoming Baseline Limitations: We completely restructured the project into a production-ready MLOps framework.

1. **Modular MLOps Architecture:** Adhered to DRY principles, decoupling data engineering, training, and evaluation into reusable Python modules.
2. **Rich Exogenous Signals:** Injected pricing elasticity, UK holidays, and cyclical temporal encodings.
3. **Cluster-Aware Modeling:** Replaced the noisy, computationally expensive per-SKU approach with robust global cluster models.
4. **Advanced Architectural Diversity:** Expanded the library to include LightGBM, Ridge, and the NS-Transformer.
5. **LLM Semantic Orchestration:** Replaced Regex parsing with 768-D latent embeddings via the Google Gemini API.



Improvements & Next Steps: Computational Efficiency

The transition from a per-SKU training loop to a **Cluster-Based** architecture redefined the pipeline's operational efficiency.

Previous Pipeline (Per-SKU)

- ▶ Trained 1,000 independent models iteratively.
- ▶ Ignored shared seasonal patterns.
- ▶ **Execution Time:** ≈ 60 Minutes.

Our MLOps Pipeline (Cluster-Aware)

- ▶ Trains 5 global models on aggregated data.
- ▶ Includes complex SOTA Transformers.
- ▶ **Execution Time:** < 5 Minutes.

Efficiency Gain: 91.67% Reduction in
Computational Load



Improvements & Next Steps: Head-to-Head Benchmarking

Despite the inherent difficulty of predicting aggregated cluster shapes versus fitting individual SKU noise, our feature-rich models demonstrate superior generalization.

Model Architecture	Current Project (Ours)		Previous Group	
	WMAPE	Med. MAPE	WMAPE	Med. MAPE
LightGBM	57.13%	52.46%	74.24%	65.66%
NS-Transformer (SOTA)	60.81%	55.86%	N/A	N/A
Ridge Regression	60.99%	56.54%	N/A	N/A
SARIMAX	90.28%	79.61%	61.79%	57.24%
Prophet	89.37%	78.46%	N/A	N/A
Global Benchmark (Avg)	60.99%	56.54%	71.36%	65.52%

- **Key Takeaway:** The inclusion of exogenous features and semantic context more than compensates for the loss of individual SKU-level granularity.



Improvements & Next Steps: Recommendation

- ▶ **Hybrid Cluster-Aware Strategy:** Move away from unstable per-SKU models in favor of a cluster-based architecture. This approach maximizes the signal-to-noise ratio and ensures mathematical stability across the long-tail catalog.
- ▶ **Agentic Decision Support:** Fully deploy the LangGraph-based assistant to transform raw numerical forecasts into actionable supply chain insights, effectively bridging the gap between data science and warehouse operations.
- ▶ **Production Readiness:** The modular nature of the new codebase and serialized model artifacts allow for immediate integration into existing ERP systems via a unified API.



Improvements & Next Steps: Strategic Roadmap

To further evolve the predictive accuracy and operational utility of the system, we propose four key research directions:

- ▶ **Hierarchical Forecast Reconciliation:** Implement a Minty or Top-Down strategy to align our cluster-level forecasts with exact individual SKU-level constraints.
- ▶ **Inventory-Aware Objective Functions:** Modify loss functions to prioritize the minimization of *Under-Forecasting* (stock-outs) over *Over-Forecasting* (holding costs).
- ▶ **Real-Time Demand Sensing:** Integrate the Agentic AI layer with real-time web traffic or social media APIs to adjust the 12-week forecasts based on intra-week behavioral shifts.
- ▶ **Automated Ensembling (Stacking):** Develop a meta-model that dynamically weights LightGBM, SARIMAX, and NST based on the specific uncertainty interval of each SKU.



The Team



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Thank you!

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